

Rabbit and Rodent Dentistry

Hugues Beaufrère

Dental disease is one of the most common diseases of rabbits and rodents and lead to many secondary conditions. It is very common to manage these diseases long-term in these patients so a thorough understanding of dental clinical anatomy, dentistry work up, and treatment is important.

Rabbits and many rodents have evolved to eat large quantities of **fibrous, abrasive, and energy-low food** (otherwise known as grass) partly by having many dental adaptations. However, the provision of certain captive diets in combination with many other factors (many associated with selective breeding) may promote the occurrence of dental malocclusion.

The management of dental disease in rabbits and rodents will depend on your ability to correctly identify dental and dental-associated (=odontogenic) lesions and apply adequate dentistry diagnostic and treatment techniques.

We won't cover the dentistry of non-herbivorous small mammals in VET436 as their dentistry is very similar to domestic carnivores, so you can more readily apply your general dentistry knowledge to these species. However, dentistry techniques and approaches are very different in rabbits and rodents and for this reason, they have a dedicated lecture.

CLINICAL ANATOMY

General Dental Anatomy

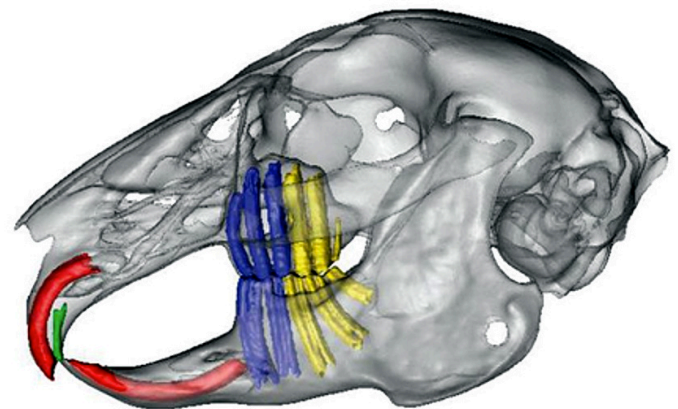
Remember from your amazing "clinical anatomy" lecture that rabbits and rodents are only distantly related and therefore will have marked (and clinically relevant) differences in their dental anatomy and physiology.

The first thing to know about the teeth in herbivorous small mammals is their **dental formula**. While the dental formula varies by species, fortunately, they are easy to remember as only the incisors and premolar numbers

change. See below:

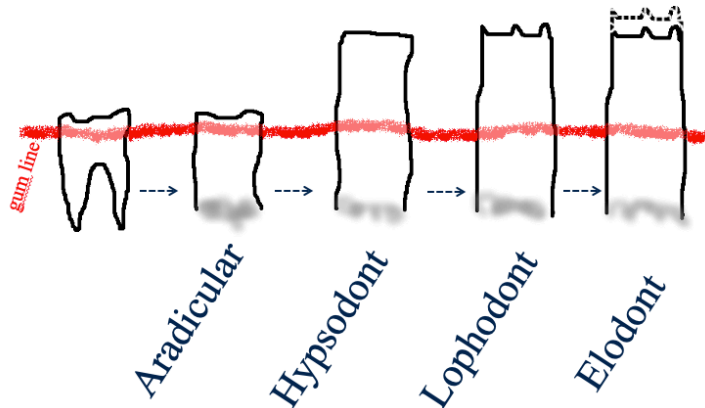
SPECIES	I	C	PM	M
Rabbit	2/1	0/0	3/2	3/3
Hystricomorph rodents	1/1	0/0	1/1	3/3
Myomorph rodents	1/1	0/0	0/0	3/3
Sciurimorph rodents	1/1	0/0	2/1	3/3

In terms of dental nomenclature, the maxillary incisors are dI2 (d for deciduous, which is retained in rabbits) and I3, the mandibular incisors are dI2 (nomenclature comes from this [article](#) in case you are wondering), **the first maxillar premolar is P2 and the first mandibular premolar is P3** (because of vestigial/deciduous teeth). The second maxillary incisors (I3) are very tiny and so is the last maxillary molars (M3). There is a large space between the incisors and the molars, called the diastema (or interdental space). **In guinea pigs and chinchillas, the only premolar is P4.**



Rabbits and hystricomorph rodents have aradicular (no roots), elodont (continuously growing), hypsodont (high clinical crown) and lophodont (transverse enamel ridges to grind food, creating dental subunits in some species). The terms aradicular, hypsodont and elodont can also be combined into

the term hypselodont teeth. These teeth are divided into the clinical crown (what is above the gum line) and the reserve crown (what is below the gum line). Since there is no root, you should never refer to teeth roots in these species (such as tooth root abscess, elongated roots and so on). As mentioned above, these teeth are continuously growing and the rate of growth is variable (8-10 mm/month for incisors, 4 mm/month for cheek teeth) and may depend on other factors such as occlusal pressure and diet).

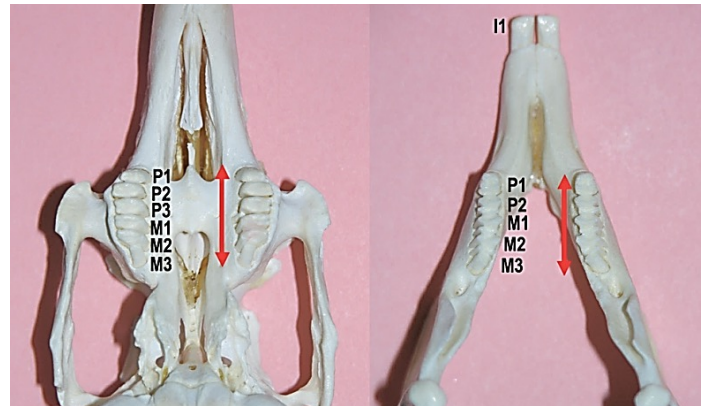
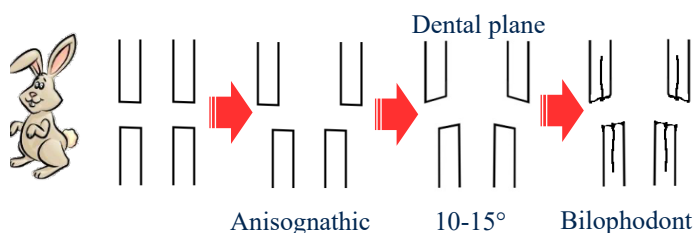


In contrast to rabbits and hystricomorph rodents, myomorph (rat-like) and sciurimorph (squirrel-like) rodents only have incisors that are hypselodont. **Their cheek teeth are actually brachydont (short clinical crowns), anelodont (not continuously growing) and bunodont (occlusal surface with rounded enamel cusps).** So you can only trim their incisors, but not their cheek teeth.

While the occlusal plane of the cheek teeth vary by species, the **chisel-shaped occlusal plane of the incisors** is present in rabbits and most rodents.

Rabbit Peculiarities

In rabbits, all the cheek teeth (except the maxillary P2 and M3) are **bilophodonts** (they have 2 subunits, 2 pulp cavities). Their dental arcades are also **anisognathic** with the upper arcades being wider than the lower arcades. The dental plane is **slightly oblique at 10-15 degrees**. Also the teeth are in a straight line from caudal to rostral.



In normal occlusion, **the mandibular incisors occlude against the peg teeth** (second maxillary incisors, I3, see figure on page 1). **Each cheek tooth occludes with 2 other cheek teeth except for the simplified maxillary P2 and M3.** Another intriguing fact is that the mandibular P3 is much bigger and tougher than the other cheek teeth and is interestingly the most commonly affected by crown elongation. **The occlusal surface of rabbits is in zigzag and is not flat.**



Rabbits chewing pattern is latero-lateral, which, in malocclusion, will frequently translate into sharp dental points.

Another thing that is highly clinical relevant in rabbits is the topographical relationship of the maxillary teeth with other cranial structures. Why is it important you may ask? Because with retrograde elongation of reserve crowns, these adjacent structures will be impacted so you will have to focus your attention on specific cheek teeth depending on the issue. **The maxillary incisors and P2 (first maxillary P) are closely associated with the nasolacrimal duct and reserve crown elongation may lead to occlusion of this duct.** In addition, the **upper P2 is adjacent to the nasal cavity and sinus. Its elongation is the main cause of odontogenic rhinitis in rabbits** (see CT image below).



P3 is more associated with the zygomatic arch and its elongation may be palpable during physical examination if long enough.



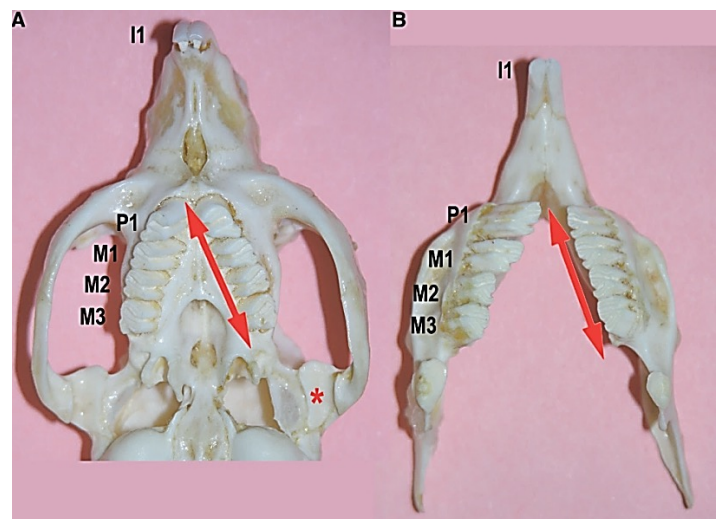
The apex of the more caudal cheek teeth (P4 and M1-3) are immediately below the orbit and their reserve crown elongation may lead to retrobulbar abscesses.



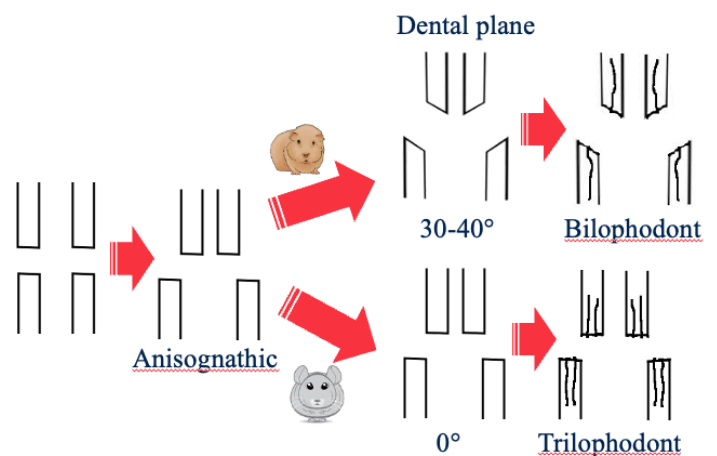
Guinea Pig and Chinchilla Peculiarities

In hystricomorph rodents, the teeth are also polylophodont. They are **bilophodont** in guinea pigs and **trilophodont** in chinchillas (and degus). The **anisognathism** is the opposite of rabbits in that the lower arcades are wider than the top arcades. In normal occlusion the incisors do not touch. In most rodents, except guinea pigs, the mesial aspect of the incisors are yellow to dark orange due to ferrous deposits.

The teeth are also in an oblique alignment on the dental arcades rostro-caudally. You can see that when the first mandibular premolars get elongated, they can bridge, which can lead to tongue entrapment and anorexia.



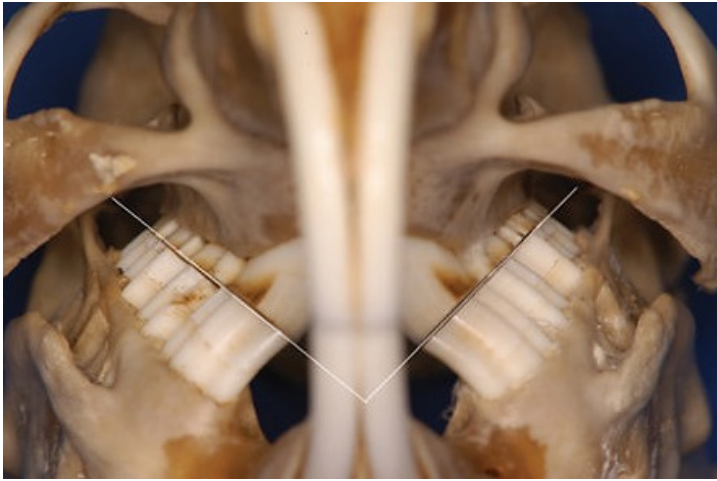
The dental plane is very oblique in guinea pigs (30-40 degrees) and flat in chinchillas (0 degrees). The occlusal surface tends to be flat unlike rabbits. See below for state-of-the-art medical illustrations on that.



Guinea pigs and chinchillas chewing pattern is **diagonal-elliptical**. For this reason, in malocclusion, they tend to have more homogeneous dental alterations and

have less points than rabbits.

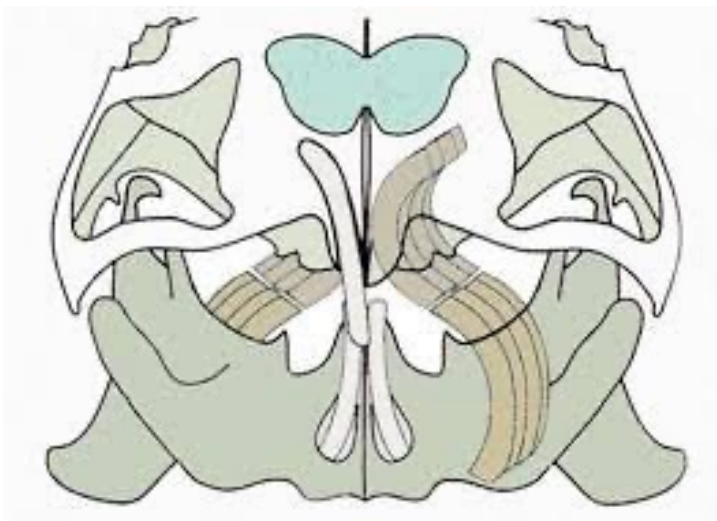
Guinea pig cheek teeth occlusion (frontal view):



Chinchilla cheek teeth occlusion (frontal view):



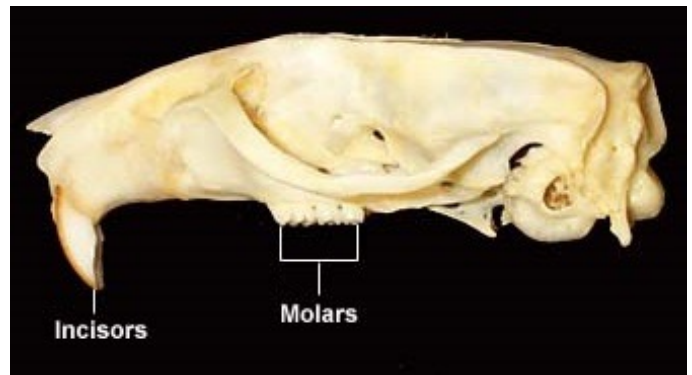
The other thing is that, in guinea pigs, **the reserve crowns are extremely curved**, which makes them extremely hard to extract.



Rat and Mouse Peculiarities

We'll spend less time on rats and mice (and gerbils and hamsters), because they ~~are less interesting~~ have less dental disease (except incisor malocclusion). The incisors are huge, hypselodont and with yellow to dark orange enamel. These rodents need materials to gnaw to keep the incisors at a physiological length. As in most rodents, the incisors do not touch in normal occlusion when the mouth is closed. **The mandibular incisors can be spread at the tip**, which should not be confused for loose teeth.

The cheek teeth (only 3 molars on each quadrant) are anelodont and the jaws are isognathic.



DENTAL DISEASE - DIAGNOSIS

Introduction

Dental malocclusion and associated changes occur very commonly in pet rabbits and rodents, more so than in laboratory animals (which don't live very long). Several risk factors are likely associated with the development of dental malocclusion such as metabolic bone disease, genetic factors, skull conformation issues (short-face rabbits), and age. In guinea pigs, deficiency in vitamin C may also predispose to the disease. An inappropriate diet may also lead to variable dental quality and degree of attrition. Any conditions that induce a change in chewing pattern, dental wear, and occlusal pressure could ultimately lead to pathologic dental changes. Most rabbits tend to develop acquired dental disease after 5 years old.

What we include under the umbrella term of dental disease is actually a **wide variety of dental lesions** that may include dental malocclusion, elongation of clinical crowns, elongation of reserve crowns, change in dental plane, change in tooth curvature, presence of dental spurs and associated lingual or buccal ulcers, presence of infection, abscess, and cavities, the occurrence of periodontal lesions, alveolar bone remodeling, and

temporomandibular joint disease. You can see that a diagnosis of dental disease is not going to cut it, you need to characterize it a bit more as well as look for dental-associated conditions. **The treatment will depend on the correct identification of the problem and is typically life-long.** It's also more fun to know what you are doing and treat in a targeted manner.

Unfortunately, dental disease has tremendous secondary effects on other organ systems and adjacent head structures.

Ophthalmic: occlusion of nasolacrimal ducts (by incisors and upper P2), dacryocystitis, retroorbital abscesses (upper P4 and molars).

Respiratory: Rhinitis, nasal empyema, abscess, and dyspnea all frequently caused by upper P2.

Gastrointestinal: dental pain and malocclusion may lead to pain and hyporexia, which in turns may lead selective feeding and a decrease intake of non-digestible fibers, which may lead to gastrointestinal stasis and dysbiosis. Incisor issues and pain may also lead to decreased cecotrophy.

Musculoskeletal: odontogenic abscesses and odontogenic rhinitis may lead to tremendous bone remodeling and facial osteomyelitis which can disrupt adjacent anatomical structures and spread hematogenously (such as with a thoracic or abdominal abscess).

Cutaneous: rabbits with incisor malocclusion may not self-groom well, which can lead to unkempt coat. Excessive salivation (more common in chinchillas) may also lead to moist chin dermatitis.

History and Physical Examination

The reasons of presentation of rabbits and rodents with dental disease frequently include non-specific presentations such as **decreased activity level, anorexia/dysorexia, or lack of stool production.** Specific reasons of presentation may include the presence of a **facial mass, facial abscess, epiphora, bruxism/abnormal mastication, or nasal discharge with dyspnea and sneezing.** On history, you may find evidence of **selective feeding** (only soft food, only pellets, decreased hay consumption) and an **inappropriate diet** being offered (energy rich low abrasive diet).

There are several clinical signs you may evidence on the physical examination. In rabbits, you can palpate the jaw (especially the inner part as elongated reserve crowns tend to grow a bit medially) and try to feel bumps and lumps. Palpation of the zygomatic area may reveal a bump caused by an elongated upper P3. A moist chin or dermatitis may be caused by hypersalivation. It

is uncommon in rabbits except with oral ulceration.



However, it is particularly common in chinchillas and can be seen in guinea pigs too. Epiphora and conjunctivitis should raise the suspicion of an occlusion of the lacrimal duct (upper incisors and P2).



Nasal discharge or dyspnea should raise the suspicion of an odontogenic rhinitis.



Exophthalmia with a negative retropulsion may indi-

cate the presence of a retrobulbar abscess (upper P4 and M1-3) and may be further complicated by an exposure keratitis or corneal ulcer.



When done with the general physical examination, you should do a **cursory dental examination**. You can look at the incisors very easily by spreading the lips. For the cheek teeth, you can look at them with either an oral speculum or a large ear cone. You won't be able to detect a lot of subtle changes or abnormalities affecting the back molars, but this is still extremely valuable to have a quick assessment of the dentition.



If you suspect dental disease, then a dental work up should be done and typically include some imaging (skull xrays or CT-scan) to check the reserve crowns and a more in-depth dental examination under anesthesia to check the clinical crown, typically with an endoscope (stomatoscopy).

Skull Radiographs

While skull CT-scan is ideal, just because of the complex topographical relationships of the teeth with adjacent head anatomical structures, you can still see a lot on radiographs. A good set of skull radiographs include a **straight lateral (ear bulla superimposed), a DV (make sure the head is flat against the radiography table), 2 obliques to isolate dental arcades, and a skyline to see the TMJ**. The most useful view is the

lateral and we frequently use **“reference lines”** to assess teeth length and the occlusal plane.

In rabbits, the **first reference line is for the occlusal plane** and goes from the tip of the hard palate to where the bulla meet the base of the skull. If the occlusal plane is not on that line, it typically means it is shifted or abnormal. The **second reference line for the reserve crown of the maxillary cheek teeth** goes from the tip of the nasal bone to the tip of the occipital bone. Anything above that line is an overgrown reserve crown. **The last two lines follow the hard palate and the floor of the interdental space**. They should be parallel or converge or it means that the mouth is forced open by the elongated clinical crowns.

In guinea pigs, the first reference line goes from a point situated at the projection of the floor of the interdental space at the incisors to the top of the bulla. The second reference line goes from the tip of the nasal bone to the top of the bulla. The last 2 lines, the palatal and mandibular plates, should strongly converge in guinea pigs. Note that many guinea pigs have a mild elongation of incisors, so they can still be found to be in contact.

In chinchillas, these reference lines are less intuitive. The occlusal plane reference line goes from the same rostral landmark as the guinea pig to a third of the bulla, the reserve crown one goes from the top of the incisor to half of the bulla, and the 2 last ones are the same as in rabbits and guinea pigs.

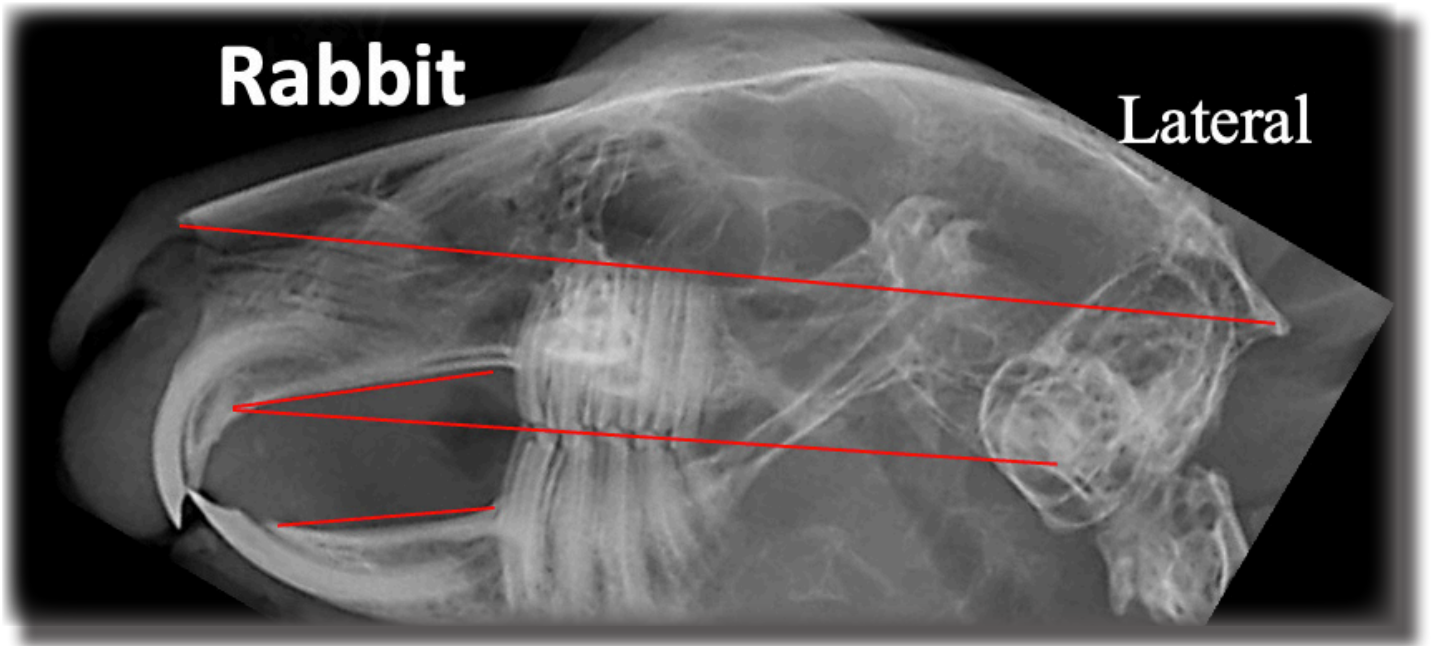
On top of the reference lines, you should assess the overall dentition. Look at the mandible and whether it looks bumpy or not with retrograde elongation of mandibular reserve crowns. This will be particularly noticeable on the oblique views. **Look at the periodontal space, the length of clinical and reserve crowns, the presence of bone remodelling, the shape of the occlusal plane and the structure of the teeth**.

In chinchillas and guinea pigs, the elongation of clinical crowns frequently lead to a mild mandibular prognathism so the first mandibular premolar is often more rostral than the top one.

You can see the reference lines on lateral xrays of the rabbit, guinea pig, and chinchilla on the next page followed by some examples of radiographic dental abnormalities.

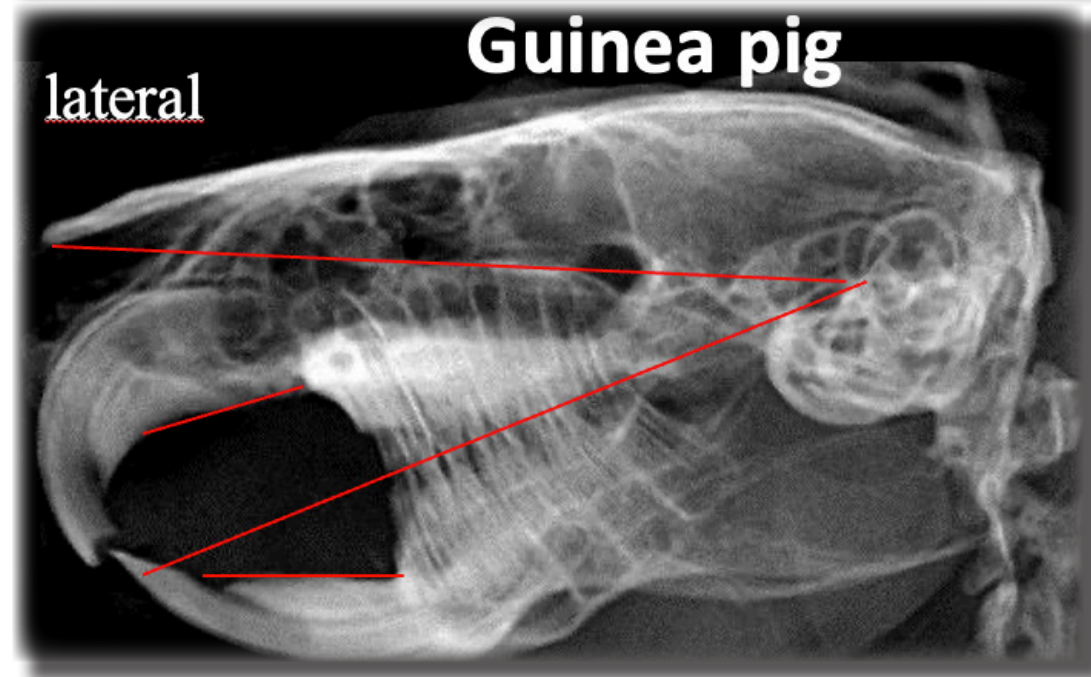
Rabbit

Lateral

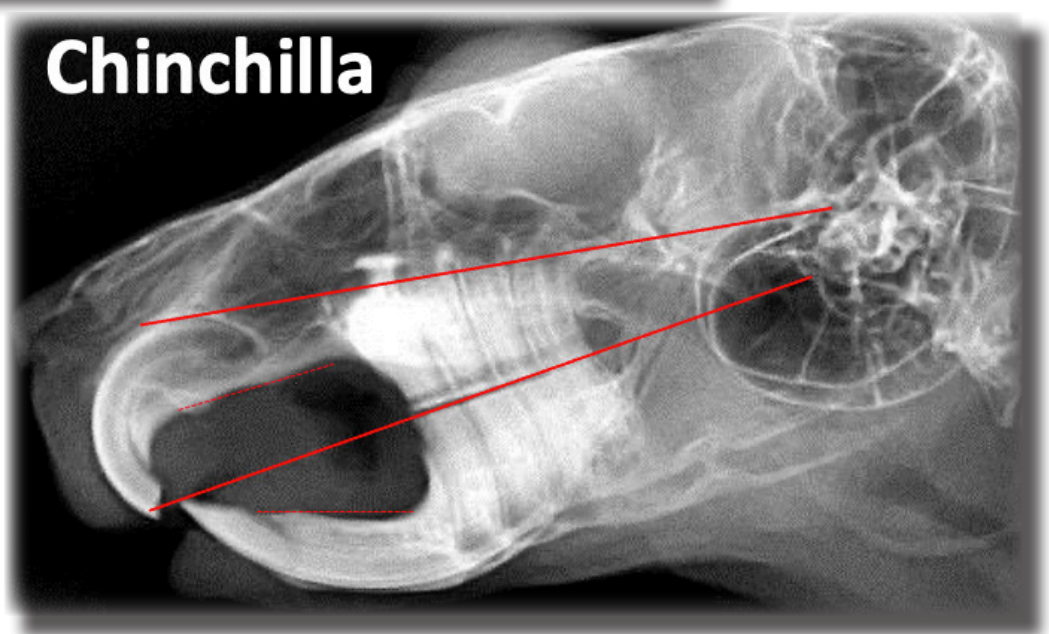


Guinea pig

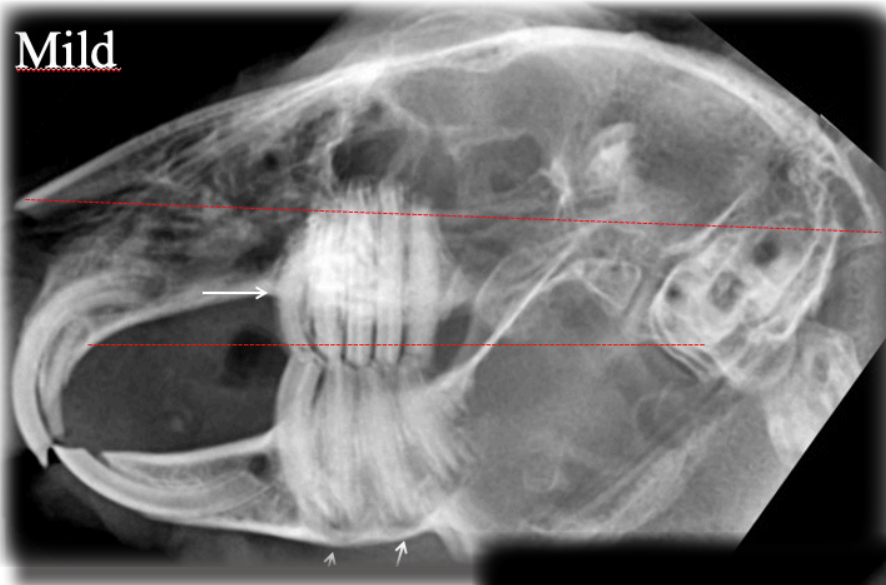
lateral



Chinchilla



Mild



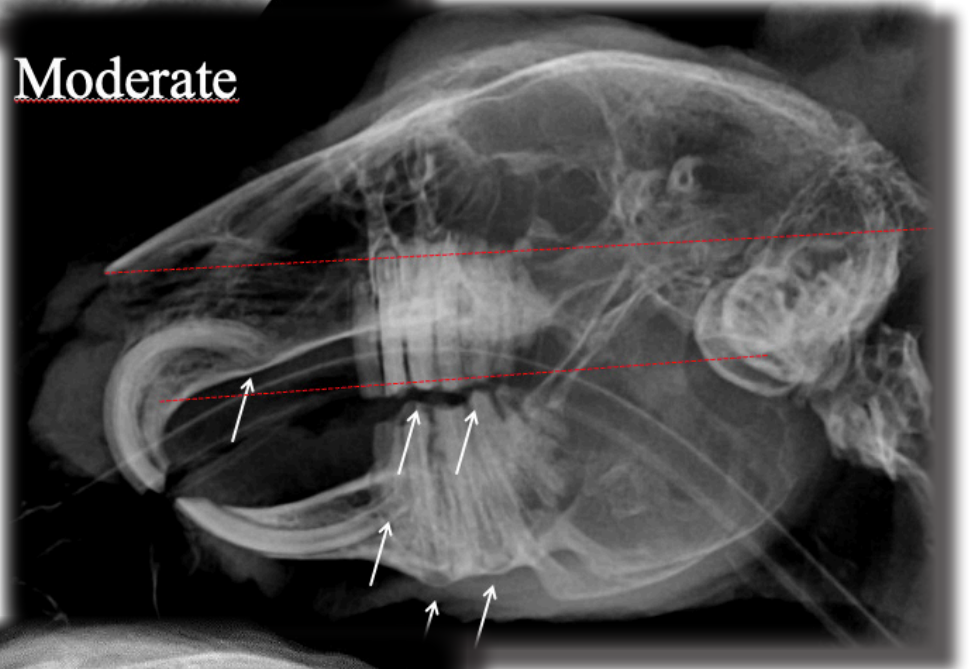
Rabbit, lateral view

These mild changes are characterized by a mild elongation of reserve crowns of the maxillary (above reference line) and mandibular (arrows). There is also a mild angulation of some teeth, elongation of clinical crowns (especially mandibular P3) and an uneven occlusal surface, which is taller rostrally. There is increased space between the lamina dura and the upper P2 indicating some degrees of periodontal disease.

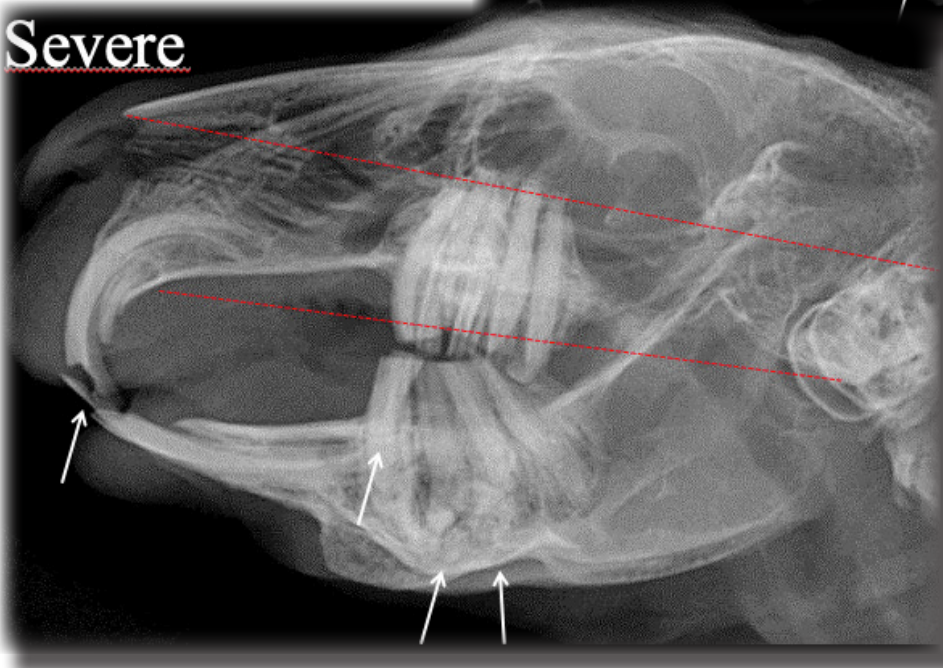
Moderate

Rabbit, lateral view

These moderate changes are characterized by further elongation of the clinical and reserve crowns of most teeth (especially the incisors and mandibular cheek teeth). The occlusal surface is very uneven showing some steps. The mandible starts showing some boney remodelling.

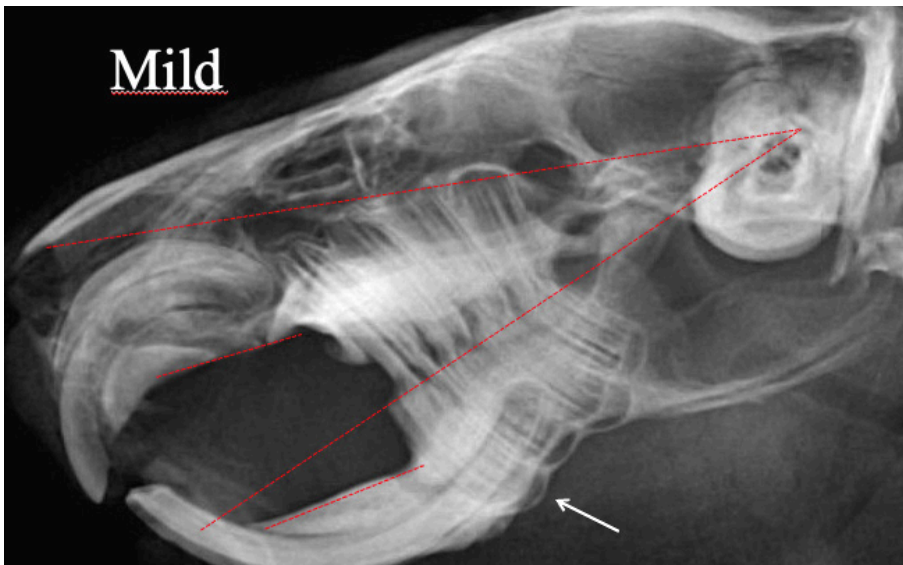


Severe



Rabbit, lateral view

These severe changes are characterized by more striking incisor and cheek teeth malocclusion, more angulation of teeth, disruption of the occlusal plane and drastic remodelling of the mandible.



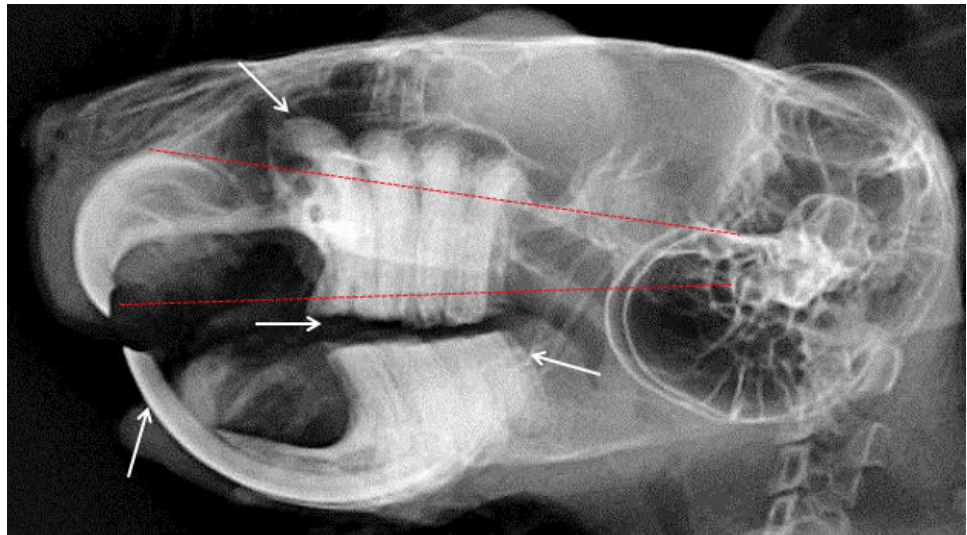
Mild

Guinea pig, lateral view

These mild changes are characterized by mild elongation of both clinical and reserve crowns of incisors and cheek teeth leading to bumpiness of the ventral aspect of the mandible, mild prognathism of the mandible and incisor malocclusion.

Chinchilla, lateral view

Most chinchillas tend to already have moderate-to-severe dental disease by the time they are diagnosed. In this one, there are severe reserve crown elongations of the maxillary cheek teeth going passed the reference line. There are also severe elongation of the clinical crowns with a disrupted occlusal plane and the mouth is forced open by the long teeth. The back molars also act as a fulcrum. The incisors are also elongated.



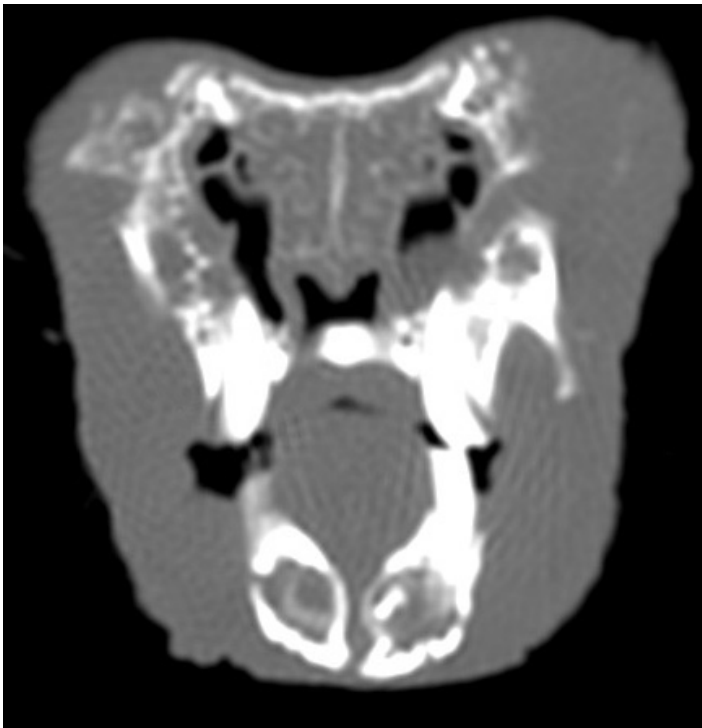
Skull Computed Tomography Scan

Skull CT or cone-beam CT (higher resolution for bones, but not good for soft-tissue, frequently used for dental CT) are extremely valuable in the diagnosis of the multiple dental and extra-dental alterations occurring in rabbits and rodents' dental disease. We won't spend too much time on it as it is less available in private practices than standard radiographs. However, it is still important to understand the value of it and how it can help you treat these patients. It is particularly useful for the following conditions/applications:

- **Characterizing odontogenic abscesses**, which teeth are involved and the topographical relationship with adjacent structures. CT contrast also allows to easily delineate the abscess capsule. Some amount of gas in there is also common with dental abscesses. It is especially useful for retroorbital abscesses.



- **Diagnosing and staging odontogenic rhinitis** in rabbits. It is very common and CT helps to see which teeth are involved (typically upper P2) and may need to get extracted. It is also useful to plan surgical approaches (rhinotomy).



- **To better overall stage dental disease and better monitor progression of dental lesions.** There is a [nice article](#) from UC Davis on CT findings in rabbit dental disease if you are interested.



Stomatoscopy

Because the oral cavity of rabbits and rodents is small and does not open much, it is extremely difficult to have a thorough diagnosis of all clinical crown abnormalities with the physical examination alone on an awake animal.

Stomatoscopy allows to perform a more complete examination on an anesthetized animals with the following advantages:

- Focused lighting

- **Magnification**

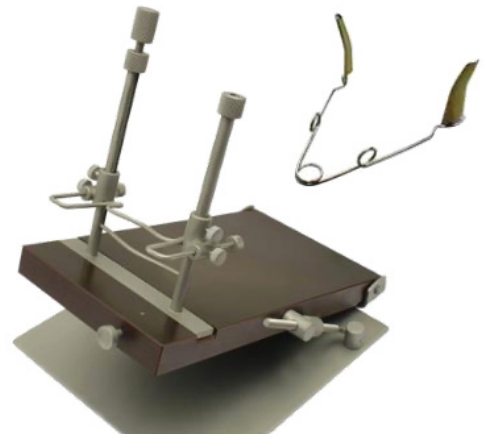
- Angled view to visualize difficult areas

- Photo and video recording for documentation and showing lesions to owners.

Different **rigid endoscopes** can be used, but typically a short 0 or 30 degree angle scope.

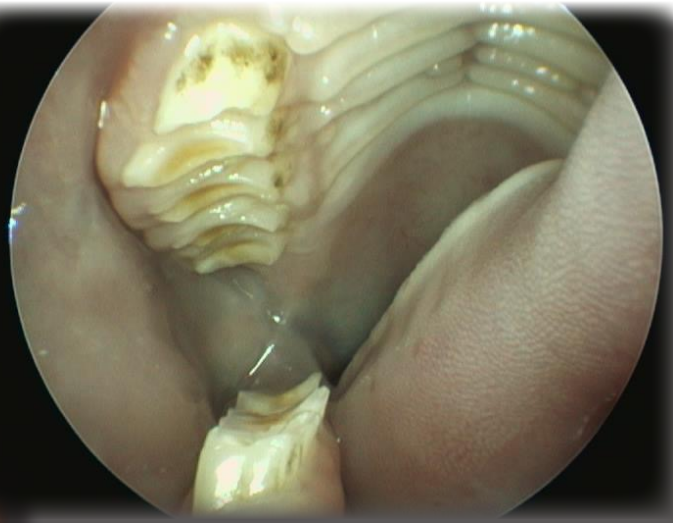


The rabbit or rodent is anesthetized and placed on a **dental table** (made for rabbits and rodents with an adjustable mouth gag and table angle). A **cheek spreader** is then placed in the mouth.



The examination is performed while looking on a screen. The overall mouth is first assessed paying particular attention to dental arcade asymmetry, disruption of the occlusal surface and angle (remember the normal occlusal angles of rabbits of 10-15 degrees, guinea pigs of 30-40 degrees, and chinchillas of 0 degree). Then, individual tooth are inspected for shape, the presence of abnormalities such as point, rotation, elongation, food pockets, and mobility. Please see below for some examples of abnormalities diagnosed on stomatoscopy.

Stomatoscopy on a rabbit

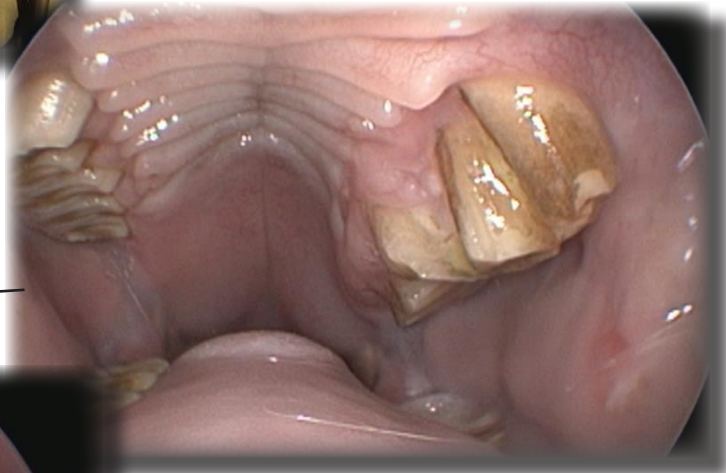


Rabbit - normal

Note the normal(ish) occlusal angle, the absence of points, angulation, ulcer, food pockets, and obvious clinical crown elongation. Because of the 10-15 degree angle of the occlusal surface, the bottom cheek teeth always appear to have mild points. There may be a bit of gingival hyperplasia on this rabbit.

Rabbit

There is obvious asymmetry with the upper left arcade being abnormal. There are marked clinical crown elongation, a disrupted occlusal angle and gingival hyperplasia.



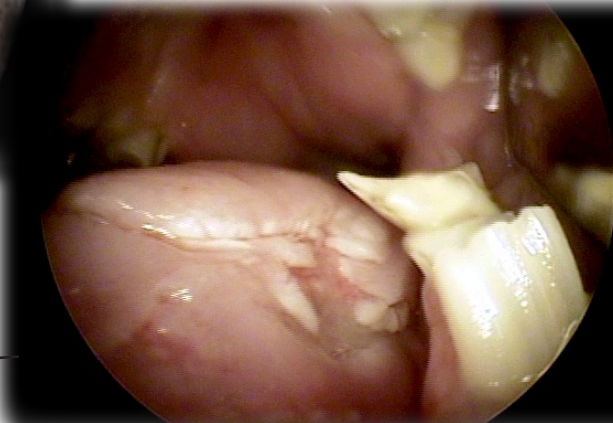
Rabbit

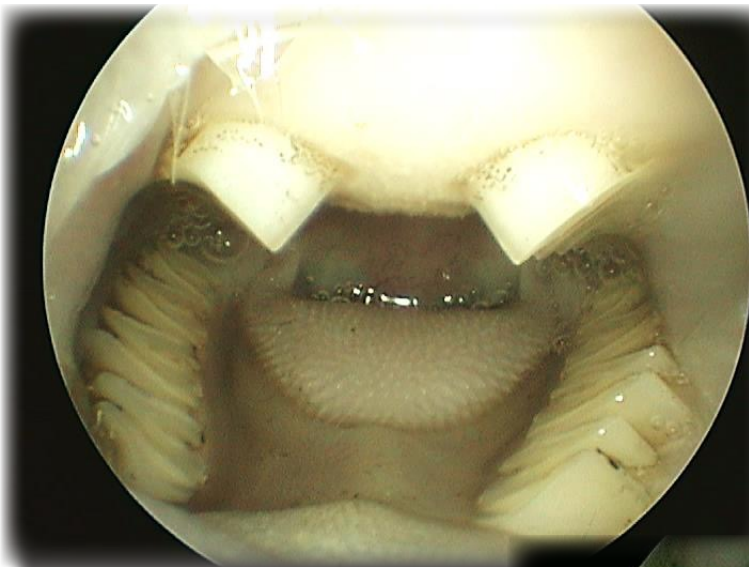
There is also a dental asymmetry with disrupted occlusal surface and elongated clinical crowns. In addition, there are sharp points on upper P2-4, which caused a buccal ulcer (hidden by the cheek spreader). Maxillary cheek teeth often cause buccal spurs, because of how the occlusal angle is.



Rabbit

While upper cheek teeth tend to have buccal points and ulcers in rabbits, it is the opposite for the mandibular cheek teeth. Here is an example with lingual points and a tongue ulcer.



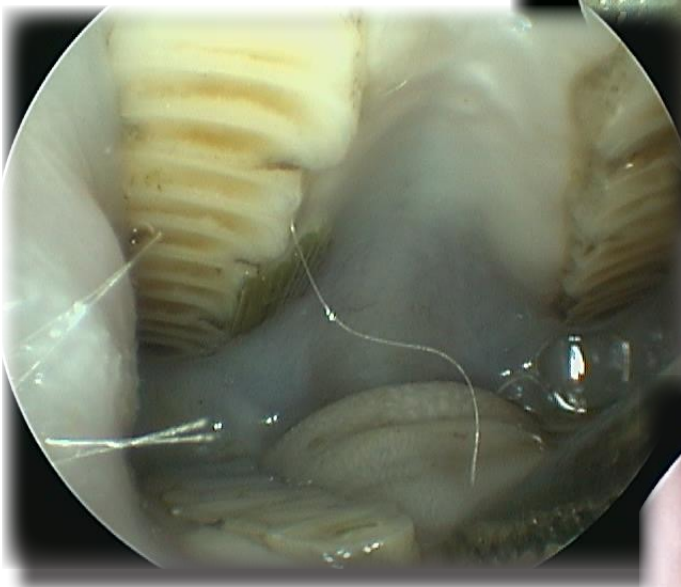


Guinea pig - normal

Normal occlusion of the cheek teeth in guinea pigs. You can see that the occlusal angle is very sharp at about 30-40 degrees. The anisognathism is also easy to appreciate. Guinea pigs have the particularity to salivate a lot and often have food in their oral cavity.

Guinea pig

Here is the infamous tooth bridging and tongue entrapment specific to guinea pigs. Because teeth converged rostrally and mild prognathism occurs with cheek teeth clinical crown elongation, the result is often a drastic elongation of mandibular premolars.

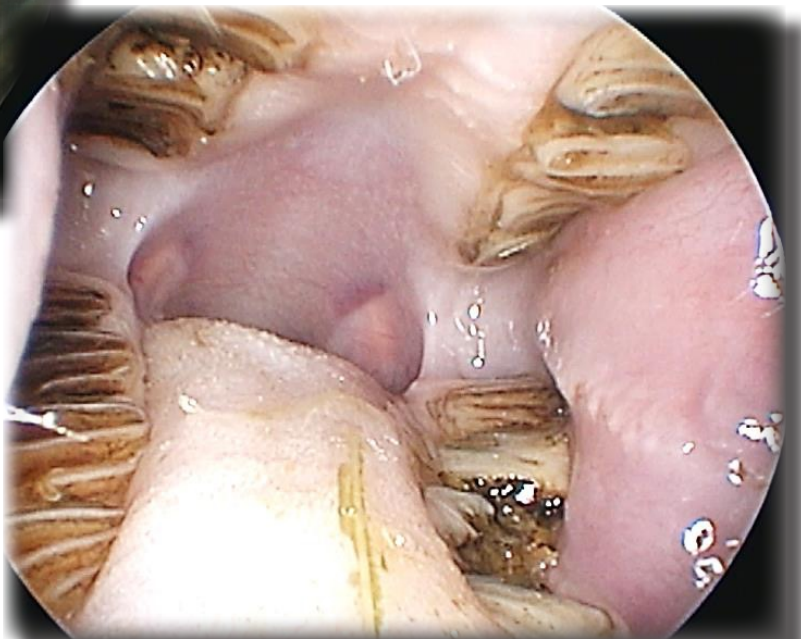


Chinchilla

Normal occlusion with a flat occlusal plane. You can easily see that the cheek teeth are trilophodont on this picture.

Chinchilla

Abnormal mouth. There is asymmetry in the dental arcades and some significant disruption in the occlusal plane (especially on the upper left). The teeth are darker at some spots, likely as a result of remaining plant pigments due to a decrease in attrition. There is also a cavity on M1 on the bottom left. Cavities, for some reasons, are very common in chinchillas (but are not in other species).



DENTAL DISEASE - TREATMENT

Clinical Crown Reduction and Occlusal Adjustment

You should never trim incisors or cheek teeth with nail clipper, wire cutters, molar cutter, manual rasps, or large cutting disks on dremel[®]-type units. Soft tissues are easily traumatized by these techniques, it may induce increased pain, and may lead to dental fracture, splitting, and periapical pathology.

Instead, a **dental unit with a straight handpiece and diamond-coated dental burs** should be used. A variety of brands and burr tip types can be used. When trimming teeth, you should protect the adjacent soft tissues with either a metal spatula or a **soft-tissue protector** (see below a burr placed within a straight handpiece and within a soft tissue protector).



Make sure you don't cause thermal trauma to the teeth when trimming. You may wipe the teeth with a water soaked cotton tip applicator or just trim at intervals instead of continuously.

The goals of the dental procedure are the following:

- Remove spurs** to prevent ulcer and pain

- Shorten clinical crowns** to decrease occlusal pressure, which is linked to retrograde reserve crown elongation and forcing the mouth to be more open, thereby disrupting normal chewing motions. The amount of crown reduction that is needed is a matter of debate. You can shorten clinical crown minimally to what would be normal for the species or maximally to increase the time interval in between dental procedures. In any case, you should make sure the dental pulp is not exposed, which would be uncommon, but can still happen.

- Reestablish normal occlusal plane** (remember the species-specific angles) to allow the animal to chew in a physiological manner.

- Trim cavities.** Teeth are hypselodont, so by trimming away the infected dental tissue, the tooth will regrow normally (also with antibiotics).

- Prevent reserve crown retrograde growth** by decreasing occlusal pressure.

- Remove diseased teeth intraorally** (see later).

- Clean interdental and periodontal pockets.**

- Prevent iatrogenic trauma** such as bleeding, dental trauma, and heat damage.

An example of clinical crown reduction using a diamond-coated dental burr housed within a soft tissue protector can be seen in this [video](#).

Incisor clinical crowns can be shortened using a small diamond-coated cutting disk or a small burr on an angled handpiece. Placing a tongue depressor behind the incisors allow to protect soft tissues. **Make sure you reestablish the chisel-shaped occlusal plane.** Again, normal incisor occlusion in rabbits is when the lower incisors rest on the peg teeth, and in rodents the incisors do not touch in normal occlusion when the mouth is closed.



Tooth Extraction

Teeth extraction can be tricky in rabbits because of their long reserve crowns and the small size of the skull.

It is frequently indicated to extract incisors in rabbits for congenital incisor malocclusion as periodic trimming can be too frequent and unpractical life-long. They may also need to be extracted in acquired malocclusion if associated with abscessation or abnormal direction. It is best to intubate rabbits for this procedure. Local anesthetic blocks with lidocaine can be done. An **incisor luxator** or small long scalpel blade or large needle can be used. The incisor luxator makes it easier, but may be too big for small dwarf rabbits. After cleaning the gum, the incisor luxator is successively inserted laterally and medially in an apical direction. Going back and forth followed by gentle up and down rotation should be done until the tooth is loose. Once loose, the incisor is grasped with a needle holder or long hemostat and pulled making sure to avoid leveraging and too much rotation that may result in fracturing

the tooth. For the upper incisors, caution needs to be exercised not to penetrate the sinus. **Once the incisor is extracted, the apex needs to be inspected to ensure the germinal tissue is present or it will grow back.** The incisor socket is then left to heal by secondary intention. Flushing may be contraindicated as the blood clot in there promotes healing.



While incisor extraction is possible in rodents, it is extremely difficult and it is best that you reserve this procedure for extreme cases. The incisors are bigger comparatively to their skull than in rabbits and the reserve crowns tend to be even longer. **Apicoectomy** has been done instead in large rodents (such as beavers) using an extraoral approach. Once the apical aspect of the tooth is removed, the incisor will stop growing and eventually fall off.

For cheek teeth extraction, the intraoral approach under endoscopic guidance is the easiest. Typically, only diseased teeth that are already kind of loose should be removed this way. A **rabbit cheek teeth luxator** is inserted rostral and caudal to the premolar/molar.



It can be inserted pretty deep by gentle mesiolateral motions. Once the tooth is loose, it can be grasped with a rabbit cheek teeth extracting forceps. The procedure

is essentially the same in rodents. However, in guinea pigs, as previously mentioned, the extreme curvature of the cheek teeth makes extraction very very difficult.

Odontogenic Abscesses

You may ask yourself what is special in the treatment of odontogenic abscesses in rabbits. Could you just lance, clean, flush, and place them on antibiotics? Well, no, it just does not work. **These abscesses invariably recur using a simple lancing technique owing to the fact these teeth are hypselodont and rabbit pus does not drain.**

These abscesses are also frequently **polymicrobial with a predominance of anaerobic bacteria**. So enrofloxacin would be a poor choice as the only antibiotic for instance.

There are essentially 2 techniques that have been proven to be effective.

The first technique is the packing technique. It consists of lancing the abscess, cleaning, flushing, packing with strips of antibiotic-impregnated gauze, and closing the abscess capsule. **Antibiotics that work the best are ampicillin and clindamycin.** The packing has to be changed weekly until resolution. The advantages are that it is minimal work for the owner, is simple, and can take care of deep abscesses. The disadvantages are that it requires repeated anesthesia and **the technique cannot be used when an oral fistula is present as these antibiotics can be lethal in rabbits if ingested.** Guinea pigs are also more sensitive to these antibiotics, even if given parenterally, so the risk of antibiotic-related enterotoxemia is much greater in them.



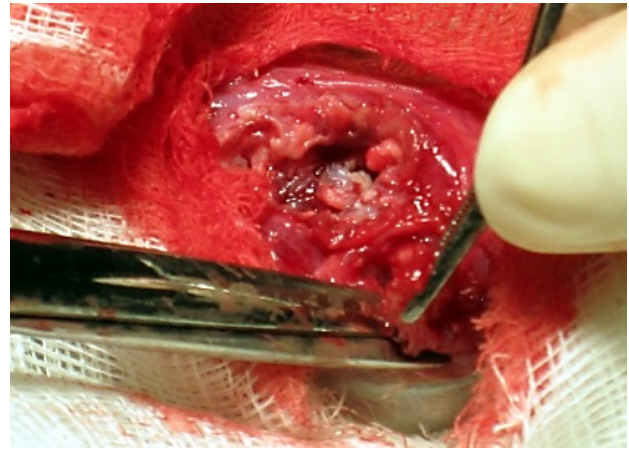
The second technique is the marsupialization technique with daily flushing. It consists in lancing the abscess and debriding as much of the infected tissues and capsule as possible. After flushing the cavity, the cavity wall is sutured to the skin circumferentially. The owner

then has to flush the resulting cavity twice a day until it closes. The advantages are that this technique is cheaper for owners and it requires a single procedure. The disadvantages are that you rely on owner compliance for appropriate flushing and monitoring and that the marsupialization site can be prone to soiling, especially on the ventral mandibular area. Some owners are just not good at restraining their pet's head and flushing deep inside these cavities.

Regardless of which technique is used, rabbits will need to also be on systemic antibiotics (with anaerobic spectrum such as azithromycin, chloramphenicol, or the combination enrofloxacin-metronidazole) and analgesics (meloxicam). Culturing the abscess is useful to base antibiotic treatment on sensitivity panels. However, anaerobic bacteria can be fastidious to grow and antibiotic sensitivity is

rarely reported for those, so be careful not to get tunnel vision on an aerobic isolate from a rabbit odontogenic abscess culture.

Tooth extraction may also be required depending on cases.



The Ten Most Important Facts

1. The dental formula of the rabbit is $2/1 \ 0/0 \ 3/2 \ 3/3$ and the dental formula of hystricomorph rodents is $1/1 \ 0/0 \ 1/1 \ 3/3$.
2. Rabbit and hystricomorph rodents (guinea pigs, chinchillas) have hypselodont (aradicular high clinical crown that continuously grow) and polylophodont (bilophodont in rabbits and guinea pigs, trilophodont in chinchillas) dentition with anisognathic dental arcades (top wider than bottom in rabbits, opposite in guinea pigs and chinchillas). Dental plane is angled in rabbits (10-15 degrees) and guinea pigs (30-40 degrees) and flat in chinchillas. Cheek teeth converge rostrally in rodents.
3. Possible risk factors for dental disease include an inappropriate diet low in fibers, genetic factors (short face for instance), age, vitamin deficiencies (D and C in guinea pigs), and disruption in chewing pattern.
4. Typical dental abnormalities include elongation of clinical crowns, retrograde elongation of reserve crowns, changes in occlusal plane, curvature of teeth, dental spurs, periapical infection and abscesses, cavities, periodontal disease, and bone remodelling.
5. Dental disease may affect other organ system by causing the following issues: occlusion of nasolacrimal duct, odontogenic rhinitis and dyspnea, gastrointestinal stasis and dysbiosis, odontogenic abscesses and facial osteomyelitis, and chin dermatitis and lack of grooming.
6. Skull palpation (especially under the mandible), inspection of the incisors, and a cursory dental examination using an oral speculum should be performed during the physical examination. However, stomatoscopy under general anesthesia is required to properly detect and diagnose all dental changes.
7. Skull radiographs should include a minimum of 4 views (lateral, DV, 2 obliques, optional skyline view). Reference lines can be used to quickly assess occlusal plane and reserve crown elongations.
8. Stomatoscopy uses short rigid endoscopy to both diagnose and treat dental abnormalities and is now considered standards of care for rabbit and rodent dentistry.
9. Treatment of dental disease mainly relies on crown reduction and occlusal adjustment using a diamond-coated burr on a straight handpiece. Incisor and cheek teeth can be extracted using rabbit specific luxators.
10. There are 2 techniques for odontogenic abscesses treatment: wound-packing technique with antibiotic impregnated gauze strips or marsupialization technique with daily flushing.